

MachBoost: Speculative Decoding from Local Context for On-Device LLMs

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Abstract

Autoregressive language model inference is slow in a very ordinary way: for most decoders, one generated token still means one more target-model step. Speculative decoding can reduce that serial work, but many practical variants require a draft model, modified model heads, or serving infrastructure that is awkward for local use. This paper explores a smaller question: when a local task is already grounded in nearby text, can that text itself draft useful continuations? MachBoost answers yes for a narrow but common class of workloads. It indexes prompt and local-context tokens, proposes n-gram continuations, and verifies them with the target model before committing any generated token. On an Apple M1 Max with 32 GB memory, using MLX and `mlx-community/Qwen3.5-0.8B-MLX-4bit`, MachBoost reaches a $1.38\times$ median speedup across 35 runs with token-for-token agreement against serial greedy decoding in every run. The gains are concentrated where they should be: policy, JSON, and code-continuation fixtures reach $1.60\text{--}1.62\times$ median speedups, while open-ended controls slow down. The system therefore includes a small benchmark-and-disable step rather than pretending the method is universal.

1 Introduction

Running LLMs on a laptop has become realistic, but it has not made autoregressive decoding disappear. A small model on Apple Silicon may fit in memory and still feel slow because the loop is serial: produce a token, update the prefix, run the model again. That bottleneck is especially visible in developer workflows, where the model often sits inside an editor, command runner, notebook, or local assistant.

Speculative decoding offers a way around part of this loop. A cheap source guesses several future tokens; the target model checks the guess; accepted tokens are emitted without paying for one full target step each. Prior work usually gets the draft from another model, extra heads, or a reference sequence. MachBoost takes the reference idea and applies it to the local machine. If the user is asking about a repo, config, note, policy, or retrieved passage, the likely continuation may already be present in that local text.

The scope is deliberately modest. MachBoost does not try to accelerate every prompt. It tries to accelerate grounded prompts without changing the target model’s greedy output. If the local text is unhelpful, the method can be slower; the implementation exposes that fact through measurement and falls back to the serial path.

This paper makes four concrete contributions:

- a local-context drafter that proposes continuations from prompt text, files, and retrieved material;

- verifier-backed generation for MLX and Hugging Face-style runtimes, including a fix for a cache-path divergence observed in MLX;
- a Python API that measures a workload before deciding whether to use the boosted path;
- an Apple Silicon evaluation with 35 MLX runs, 100% token agreement, and a $1.38\times$ median speedup overall.

2 Background and Related Work

Speculative decoding was introduced as a way to accelerate autoregressive transformer inference without changing outputs by drafting candidate tokens and verifying them in parallel with the target model [4]. Speculative sampling independently developed a related draft-and-verify method for sampled decoding and reported $2\text{--}2.5\times$ speedups in a distributed setting [3]. Subsequent work explored architecture-level or self-drafting variants. Medusa adds multiple decoding heads to predict future tokens and verifies candidate trees [2].

MachBoost is closest to methods that exploit references or prompt overlap rather than an auxiliary draft model. LLMA observes that outputs in retrieval and reference-grounded scenarios often share spans with the input reference, then verifies copied spans to preserve greedy output [9]. Prompt Lookup Decoding replaces the draft model with n-gram matching over the prompt [5]; PLD+ extends that idea by using language-model artifacts such as attention or hidden states to rank candidate spans [6]. MachBoost differs mainly in packaging and operating assumptions: it treats local files and retrieved snippets as ordinary inputs to the accelerator, keeps benchmark data machine-readable, and makes low-overlap prompts a measured fallback case rather than an error.

The implementation targets common local inference stacks. Hugging Face Transformers provides the broad model API surface for Python inference [7]. MLX is an Apple Silicon-oriented array framework and runtime used here for Mac-native evaluation [1]. Our measurements use a Qwen-family model converted for MLX; Qwen3 is the relevant public model-family report [8].

3 Problem Setting

Let M be a target autoregressive model and let x be the prompt token sequence. Greedy decoding emits

$$y_t = \arg \max_v M(v \mid x, y_{<t})$$

for each generation step t . MachBoost receives a corpus C derived from prompt text, retrieved context, local files, or user-provided strings. The goal is to reduce wall-clock latency while returning the same sequence y that greedy decoding with M would return.

This paper stays with greedy decoding. Sampling-compatible speculative decoding is possible in principle, but greedy argmax verification gives a simple correctness check: the accelerated sequence must match the serial baseline token-for-token.

4 Method

The implementation has three moving parts: a drafter, a verifier, and a decision rule for whether to use the drafter on a workload.

4.1 Drafting from Nearby Text

The drafter indexes token n-grams from a corpus C . At generation time it looks at the current prefix $p = x \parallel \hat{y}$, searches for suffixes of p that occur in C , and proposes the tokens that followed the best matching span. Longer suffix matches are preferred, with draft length as a secondary signal. The design is intentionally plain: no learned draft model, no training step, and no hidden state from the target model.

4.2 Verification Against the Target Model

Given a prefix p and draft $d = (d_1, \dots, d_k)$, the verifier checks how many draft tokens are equal to the target model’s greedy continuation. For verifier-capable runtimes, this can be done by scoring $p \parallel d$ in a single target call and comparing target argmax tokens at the relevant positions. If the first a draft tokens are accepted, they are committed. If a residual target token is available at the first mismatch, it is emitted to preserve progress. Otherwise the system falls back to a normal target step.

The invariant is:

$$\text{Boost}(M, x, C, T) = \text{Greedy}(M, x, T)$$

for a maximum generation length T . In the implementation, every benchmark row records an `output_match` flag by comparing boosted and baseline token IDs directly.

4.3 Keeping the Cache Path Stable

One engineering detail turned out to matter more than expected. In MLX, extending a prompt cache token-by-token is not always numerically identical to rebuilding a long prefix in a single call. During testing, a non-trimmable cache could partially accept a draft, rebuild after rejection, and later choose a different argmax than the serial cache path.

MachBoost handles this by verifying drafts on a cloned trial cache when the runtime supports cloning. Fully accepted drafts commit the trial cache. On partial rejection, if the trial cache cannot be trimmed safely, only the accepted tokens are advanced on the live cache. The rejected tail is never allowed to pollute the future prefix state.

4.4 Deciding When to Use the Drafter

Drafting from local text is sometimes the wrong choice. Open-ended prompts often accept only a few draft tokens, so verification overhead dominates. MachBoost therefore exposes a benchmark step. For a prompt family, it measures:

- token match between baseline and boosted output,
- wall-clock speedup,
- accepted draft tokens divided by generated tokens,
- target-call or forward-call reduction.

The high-level API can then calibrate an accelerator:

```
from machboost import Accelerator, GatePolicy

boost = Accelerator.from_mlx(model, context_paths=["./docs", "./src"])
```

```

calibration = boost.calibrate(
    prompts,
    max_tokens=32,
    gate_policy=GatePolicy(min_speedup=1.05,
                           min_acceptance_rate=0.10),
)
text, stats = boost.generate(prompt, max_tokens=128)

```

If the measured workload fails the policy, `generate` uses the serial baseline path. This is not as exciting as always-on acceleration, but it is much safer for a package people may drop into arbitrary local workflows.

5 Implementation

MachBoost is implemented as a small Python package with no required runtime dependencies. Optional extras install backend-specific dependencies:

- `machboost[mlx]` for MLX and `mlx-lm`,
- `machboost[hf]` for PyTorch and Transformers,
- Ollama HTTP support for benchmarking and capability detection.

The core package defines a `StepService` protocol with `next_token(prefix)` and an optional `verify(prefix, candidate)` hook. A service with only `next_token` can still run through the wrapper, but it cannot skip target work. A service with `verify` can accept several draft tokens per target call. The high-level `Accelerator` API loads MLX or Hugging Face models, reads strings, files, or directories as context, and exposes `benchmark`, `calibrate`, and `generate`. The Ollama HTTP adapter is intentionally limited: the public API does not expose token IDs, logits, KV-cache snapshots, or verifier hooks, so it is useful for benchmarking and option management but not for this form of acceleration.

6 Evaluation

6.1 Setup

We evaluate on a Mac with Apple M1 Max, 10 CPU cores, 32 GB unified memory, and macOS 26.5.2. The Python environment includes MLX 0.31.2 and `mlx-lm` 0.31.3. The evaluated model is `mlx-community/Qwen3.5-0.8B-MLX-4bit`. All results come from `results/mlx_evidence_suite_20260701.json`.

The suite contains seven fixture families with five fresh nonce-bearing repeats each: policy continuation, structured JSON continuation, retrieval-style answering, code completion, repository command quote, creative open-ended generation, and short answer. Each run generates up to 24 tokens. The positive fixtures are designed to have real local-context overlap; the creative and short-answer fixtures are negative controls.

6.2 Overall Results

Across the full suite, the boosted path matches the serial baseline in every run. Median throughput increases from 135.02 to 188.13 tokens per second, corresponding to a $1.38\times$ median speedup. The p90 speedup is $1.62\times$.

Metric	Value
Runs	35
Output match rate	100%
Median baseline speed	135.02 tok/s
Median boosted speed	188.13 tok/s
Median speedup	1.38×
Mean speedup	1.32×
P90 speedup	1.62×
Median accepted draft tokens	23.0
Median forward reduction	70.83%

Table 1: Overall MLX evidence-suite results. Output match compares token IDs between serial greedy baseline and boosted generation.

Fixture	Expectation	Match	Speedup	Base tok/s	Boost tok/s
Policy quote	Positive	100%	1.62×	125.84	203.78
JSON continuation	Positive	100%	1.61×	125.55	202.66
Code completion	Positive	100%	1.60×	125.13	201.84
Repo quote	Positive	100%	1.38×	136.52	188.13
RAG answer	Positive	100%	1.12×	135.02	151.57
Creative open-ended	Negative	100%	0.95×	157.63	148.91
Short answer	Negative	100%	0.90×	156.42	140.89

Table 2: Median per-fixture results over five repeats. Positive fixtures contain expected local-context overlap; negative fixtures are included to test gating.

6.3 Per-Workflow Results

The per-fixture view is more informative than the aggregate number. Policy, JSON, and code continuation all exceed 1.60× median speedup. Repository quote is weaker but still useful at 1.38×. Retrieval-style answering improves only 1.12×; the local text helps, but fewer tokens are accepted per verification call. The negative controls are a useful sanity check: they still match the baseline, but they are slower. Their continuations are not mostly recoverable from the provided context, so the extra verifier calls are wasted.

6.4 Interpretation

These results are not evidence for a universal speedup layer. They instead suggest that grounded local workloads are separable from open-ended ones with a small amount of measurement. That distinction matters for developer tools. A repo assistant, config editor, policy search tool, or command helper can benchmark representative prompts once and avoid using the drafter where acceptance is low.

7 Limitations

The evidence here is still early. The main benchmark uses one small MLX model on one Apple Silicon machine with 24-token generations. It does not yet answer how the method behaves on

3B/8B models, long outputs, sampled decoding, or large real repositories. The drafter is also deliberately simple; unlike PLD+, it does not rank spans with model artifacts. The Hugging Face adapter exists at the package level, but the reported suite focuses on MLX because that is where the cache work has been exercised most. Ollama over HTTP remains outside the accelerated path unless the runtime exposes a native verifier hook or a patched runner.

There is also a methodological limit. Token equality is a strong check for greedy decoding, but it says little about semantic quality under sampling. For that reason, this paper makes no quality claim. The goal is lower latency when the desired output is the same.

8 Future Work

Several next steps are straightforward. The evaluation needs larger models, longer generations, and real workspaces. The drafter could use suffix automata, prompt/document segmentation, model-artifact ranking, or retrieval over multiple corpora. The decision rule could also become online: try the drafter for a short prefix, back off when acceptance is low, and re-enable it inside structured regions. Finally, a native Ollama or llama.cpp verifier hook would make the approach available to many more local users.

9 Conclusion

MachBoost is a practical experiment in using nearby text as the draft source for speculative decoding. It is not a shortcut for every prompt. On the measured MLX suite, however, it matches serial greedy decoding across all 35 runs and reaches a $1.38\times$ median speedup overall, with $1.60\text{--}1.62\times$ speedups on the strongest grounded tasks. The package design reflects the main lesson from the results: use the drafter when local context predicts the next tokens, and fall back when it does not.

References

- [1] Apple Machine Learning Research. Mlx: An array framework for apple silicon. <https://github.com/ml-explore/mlx>, 2023. Software.
- [2] Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D. Lee, Deming Chen, and Tri Dao. Medusa: Simple llm inference acceleration framework with multiple decoding heads. *arXiv preprint arXiv:2401.10774*, 2024.
- [3] Charlie Chen, Sebastian Borgeaud, Geoffrey Irving, Jean-Baptiste Lespiau, Laurent Sifre, and John Jumper. Accelerating large language model decoding with speculative sampling. *arXiv preprint arXiv:2302.01318*, 2023.
- [4] Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 19274–19286. PMLR, 2023.
- [5] Apoorv Saxena. Prompt lookup decoding. <https://github.com/apoorvumang/prompt-lookup-decoding>, 2023. GitHub repository.
- [6] Shwetha Somasundaram, Anirudh Phukan, and Apoorv Saxena. Pld+: Accelerating llm inference by leveraging language model artifacts. *arXiv preprint arXiv:2412.01447*, 2024.

- [7] Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M. Rush. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45. Association for Computational Linguistics, 2020.
- [8] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- [9] Nan Yang, Tao Ge, Liang Wang, Binxing Jiao, Daxin Jiang, Linjun Yang, Rangan Majumder, and Furu Wei. Inference with reference: Lossless acceleration of large language models. *arXiv preprint arXiv:2304.04487*, 2023.